

OpenHPLjl RigidPipe: model

State: Q

Ports: $(H_{\text{in}}, Q_{\text{in}}), (H_{\text{out}}, Q_{\text{out}})$

$$\dot{Q} = \frac{gA}{L} (H_{\text{in}} - H_{\text{out}} - h_f(Q))$$

$$Q_{\text{in}} = Q, \quad Q_{\text{out}} = -Q$$

direct equation mapping

ModelingToolkit implementation

```
@named inlet = HydraulicPort()
@named outlet = HydraulicPort()
inlet.Q ~ Q
outlet.Q ~ -Q
D(Q) ~ (g*A/L) *
      (inlet.H-outlet.H-hf)
```

OpenHPLjl SurgeTank: model

State: H

$$A_s \dot{H} = Q_{\text{inlet}} + Q_{\text{outlet}}$$

$$H_{\text{inlet}} = H_{\text{outlet}} = H$$

direct equation mapping

ModelingToolkit implementation

```
A_s * D(H) ~ inlet.Q + outlet.Q
inlet.H ~ H
outlet.H ~ H
```